

Murray S. Bennett

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Research Summary and Interests

I am a cognitive psychologist using mathematical models, behavioural experiments, and machine learning to study perception, workload capacity, and decision making in people and teams. My work includes uncertainty estimation in AI systems and computational methods for testing cognitive theories. I aim to connect these theoretical and methodological advances with practical tools, including research on human-AI collaboration.

EDUCATION

Doctor of Philosophy in Psychology – Science

2023

University of Newcastle, Australia

Collective Workload: Human Performance in Competitive and Collaborative Tasks

Bachelor of Psychology (Honours), Class I

2018

University of Newcastle, Australia

Honours Thesis: The Mental Representation of Symbolic Numerals

RESEARCH

Postdoctoral Scholar

2024-

The Ohio State University, Department of Psychology

- Develop computational methods for communicating neural network metacognition to human users, with the goal of improving human-AI collaboration and decision-making.
- Apply advanced quantitative cognitive models (including Systems Factorial Technology and General Recognition Theory) to test processing independence and interactions in social and cognitive decision-making tasks.
- Research conducted within NSF-funded projects (SES-2237119; SES-2523723).

Postdoctoral Researcher

2022-24

UT San Antonio, Psychology Department

- Quantified human perceptual judgements and combined them with computer-vision estimates to improve classification in an offline melanoma image dataset.
- Used General Recognition Theory to examine perceptual interactions and feature integration in medical image decision-making.

Data Analyst*2021-22*

Hunter New England Health Local Health District

- Conducted weekly statistical analyses of nationwide FluTracking and COVID-19 survey data for Australian federal government reporting.
- Collaborated with international partners to design, implement, and maintain large-scale survey infrastructure for data collection in New Zealand and Argentina.
- Role held concurrently with PhD candidature.

Research Assistant*2021*

University of Newcastle, School of Biomedical Sciences and Pharmacy

- Designed, conducted, and analysed experiments examining physiological correlates of cognitive workload using the Detection Response Task.

Research Assistant*2019-22*

University of Newcastle, School of Psychology

- Provided technical and methodological support for research on cognitive processing, decision-making, and perceptual judgement.
- Designed and implemented online versions of foundational cognitive psychology experiments for research and undergraduate instruction.

Research Internship*2018-19*

University of Newcastle, Centre for Brain and Mental Health Research

- Analysed cognitive workload and flight-simulator performance data from helicopter pilots evaluating novel 3D helmet display systems.
- Examined relationships between task demands, performance, and subjective workload in applied human-factors contexts.

TEACHING

Guest Lecturer

Artificial Intelligence in Psychology

OSU 2026

- Lectured on k-means clustering and supervised student-led presentations.

Course Instructor

Introduction to Psychology

UT San Antonio 2024

- Introduction to the study of and scientific approaches to attention, sensation, perception, emotion, learning, memory, problem solving, personality, development, atypical behaviour, and social behaviour.

Honours Thesis Grader

University of Newcastle, School of Psychology

UoN 2021

- Assessed and critiqued the methodology, data analysis, and scientific writing of honours-level cognitive psychology research projects.

Lab Demonstrator

Advanced Research Methods & Design

UoN 2021

- Presenting practical content and guiding students in the hands-on application of advanced concepts in research methodology, statistics, Bayesian analysis, and experimental design.

Cognitive Psychology

UoN 2020

- Introduction and overview of fundamental cognitive psychology principles, including attention, memory, language, reasoning, and problem solving, with focus on the application of experimental methods and formal psychology report writing.

Accreditation

Associate *Centre for the Integration of Research, Teaching and Learning*, 2024

- Associate-level teaching accreditation via CIRTl by completing professional development events and demonstrating expertise in evidence-based teaching.

GRANTS & FUNDING

Enhancing Trust Calibration: An Innovative Intervention for Optimal AI-Assisted Decision Making

2025

Partner Investigator (Lead CI: Dr Quentin Gronau)

Pilot Research Scheme, College of Engineering, Science & Environment, University of Newcastle

AUD \$13,708

Research conducted within funded projects

NSF SES-2237119; NSF SES-2523723 (The Ohio State University)

AWARDS & HONORS

People's Choice, Best Presentation

2024

Virtual Mathematical Psychology Conference

Best Poster

2023

UT Health Postdoctoral Research Forum

Summer Research Internship

2019

University of Newcastle, Centre for Brain and Mental Health

Best Presentation

2018

University of Newcastle Student Research Conference

Faculty of Science Commendation List

2018

University of Newcastle

PROFESSIONAL SERVICE

Conference Session Chair

2022, 2025, 2026

Society for Mathematical Psychology (2025, 2026)
Australian Society for Mathematical Psychology (2022)

Conference Reviewing and Judging

2023–2026

Graduate abstracts and presentations judge, Hayes Advanced Research Forum (OSU)
Abstract reviewer, Southwestern Psychological Association
3-Minute Thesis judge, UT San Antonio

UNIVERSITY SERVICE

Founder and Chair, Postdoctoral Writing Club

2024

UT San Antonio

Postdoctoral Service Roles

2023–2024

Postdoctoral Appreciation Week Organising Committee, UT San Antonio
Postdoctoral Mentor of the Year Judge, UT San Antonio
Postdoctoral Association Member, UT San Antonio

Community and Governance Service

2020–2021

Board of Directors, Community Representative, Newcastle University Sport

MENTORING

Laiton G. Hedley

University of Newcastle

Undergraduate; now PhD candidate. First author on collaborative human-machine team-performance papers.

Ying-Yu Chen

UT San Antonio

Master's student; now PhD candidate. Co-author, *Applied Configural Processing* (in prep.).

Serena Browning

UT San Antonio

Undergraduate; now Master's student.

Lauren Kahn

UT San Antonio

Undergraduate; now Master's student.

PUBLICATIONS

Peer-reviewed journal articles

- Hedley, L. G., Russel, L., **Bennett, M. S.**, and Eidels, A. (In Press). Benchmarking the Benchmark: A Novel Simulation Benchmark for Team Performance. *Chinese Journal of Psychology*
- Bennett, M. S.**, Hedley, L., Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2025). Human performance in competitive and collaborative human-machine teams. *Topics in Cognitive Science*, 17(2):324–348
- Kerr, A., Carlson, S., Baldwin, Z., Innes, R., Howard, Z., **Bennett, M. S.**, Evans, T., McKenzie, T., and Dalton, C. (2025). FluTracking: Weekly online community-based surveillance of respiratory illness in Australia, 2020–2022 Report. *Communicable Diseases Intelligence*, 49
- Gronau, Q. F., **Bennett, M. S.**, Brown, S. D., Hawkins, G. E., and Eidels, A. (2023). Do choice tasks and rating scales elicit the same judgments? *Journal of Choice Modelling*, 49:100437
- Garrett, P. M., **Bennett, M. S.**, Hsieh, Y.-T., Howard, Z. L., Yang, C.-T., Little, D. R., and Eidels, A. (2022). Wheel of fortune: A cross-cultural examination of how expertise shapes the mental representations of familiar and unfamiliar numerals. *Computational Brain & Behavior*, 5(1):45–59
- John, A. R., Singh, A. K., Do, T.-T. N., Eidels, A., Nalivaiko, E., Gavgani, A. M., Brown, S. D., **Bennett, M. S.**, Lal, S., Simpson, A. M., et al. (2022). Unraveling the physiological correlates of mental workload variations in tracking and collision prediction tasks. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:770–781
- Bennett, M. S.**, Mullard, R., Adam, M. T., Steyvers, M., Brown, S. D., and Eidels, A. (2020). Going, going, gone: competitive decision-making in Dutch auctions. *Cognitive Research: Principles and Implications*, 5(1):62

Under Review

- Alaukik[†], A., **Bennett[†], M. S.**, Smith, C. T., and Kvam, P. D. (2026). A road map for assessing process independence in behavioral measures of attitudes: An SFT and GRT model analysis. *Under review at Journal of Mathematical Psychology*, https://doi.org/10.31234/osf.io/jqasm_v1
- Bennett, M. S.**, Houpt, J. W., Wenger, M. J., Townsend, J. T., and Mark, L. (2026). Scalable estimation of continuous perceptual representations from human judgments and computer vision: An application to melanoma identification. *Under review at Behavioral Research Methods*, https://doi.org/10.31234/osf.io/kgeyt_v1
- Bennett, M. S.** and Kvam, P. D. (2026). Machine metacognition improves classification performance and uncertainty quantification. *Under review at Transactions on Pattern Analysis and Machine Intelligence*, https://doi.org/10.31234/osf.io/jp83f_v1
- Fitch, A., Hedley, L., **Bennett, M. S.**, and Kvam, P. D. (2026). Real-time Cognitive Modeling using Simulation-Inversion Networks. *Under review at British Journal of Mathematical and Statistical Psychology*, https://doi.org/10.31234/osf.io/vbjd3_v1

Conference Proceedings

Hedley, L. G., **Bennett, M. S.**, Love, J., Houpt, J. W., Brown, S. D., and Eidels, A. (2023). The relationship between teaming behaviours and joint capacity of hybrid human-machine teams. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 45

Reviewing Activity

Ad hoc reviewer for: *Cognitive Research: Principles and Implications, Computational Brain and Behaviour, Topics in Cognitive Science, Communications Psychology, Journal of Mathematical Psychology, Psychology and Aging*

PRESENTATIONS

Invited

Human-AI perceptual judgements of malignant melanoma. 2024, Psychology Department, The University of Newcastle, Australia.

Human-AI collaboration. 2024, Department of Psychological Sciences at Tarleton State University.

Collaborative and competitive 'Team Spirit'. 2024, Joint meeting of the Newcastle Cognition Lab and UC Irvine's MADLAB.

Talks

Bennett, M. S., Alaukik, A., Smith, C., and Kvam, P. (2025, July 25-28). Processing independence in implicit attitudes: An SFT and GRT analysis. In *Meeting of the Society for Mathematical Psychology/ICCM*

Bennett, M. S. and Houpt, J. W. (2024, June 17-21). A GRT examination of the ABCs for melanoma identification with comparison to a deep convolutional neural network. In *Virtual Meeting of the Society for Mathematical Psychology/ICCM*

PROFESSIONAL SOCIETIES

Society for Mathematical Psychology	2019-present
Psychonomic Society	2019-present
Southwestern Psychological Association	2024

PUBLIC ENGAGEMENT

2021. Online article published in *The Conversation* (Australia): *Why the competitive spirit can take over in auctions and how you can stay in control*.